

2nd sessional exam
B.Tech. 1st semester
Subject: Biology for engineers
Girijananda Chowdhury University

1) Each question will carry 1mark

(10X1=10)

o A) The general mechanism is that an enzyme acts by:

- i) Reducing the activation energy
- ii) Increasing activation energy
- iii) Decreasing pH value
- iv) Increasing the pH value

B) Which of the following is produced with the combination of apoenzyme and coenzyme:

- i) ☒ Holoenzyme
- ii) Enzyme substrate complex
- iii) Prosthetic group
- iv) Enzyme product complex

C) Enzymes are made Up of:

- i) Fats
- ii) ☒ Proteins
- iii) Nucleic acids
- iv) Vitamins

D) A _____ is a biocatalyst that boosts the pace of a reaction without changing the reaction itself.

- i) Silicon dioxide
- ii) Aluminum oxide
- iii) ☒ Enzyme
- iv) Hydrogen peroxide

E) Enzymes are regarded as _____

- i) ☒ Biocatalysts
- ii) Messengers
- iii) Inhibitors
- iv) Antibodies

F) Proteins are made up of which of the following monomers?

- i) Nucleotides
- ii) ☒ Amino acids
- iii) Monosaccharide
- iv) Fatty acids

G) How many different types of amino acids are commonly found in proteins?

- i) 5
- ii) 10
- iii) 20
- iv) 64

H) Disulfide bonds are involved in which level of protein structure?

- i) Primary structure
- ii) Secondary structure
- iii) Tertiary structure
- iv) Quaternary structure

I) Hemoglobin is an example of which type of protein structure?

- i) Primary structure
- ii) Secondary structure
- iii) Tertiary structure
- iv) Quaternary structure

J) A protein is formed with the help of joining the monomers (amino acids) through which bond formation?

- i) The ionic bond
- ii) The acidic bond
- iii) Peptide bond
- iv) The hydrogen bond

2) Answer any four

(4x5=20)

A) What do you mean by hierarchy of Protein structure? Mention each type in a brief manner. 1+4=5

B) What is Alpha-helix? Where it is found? Explain with a suitable picture. 1+1+3=5

C) Why protein is known as diverse in function? Write a short note. 1+4=5

D) What are factors that affect the activity of an enzyme? Mention briefly. 5

E) Give a brief note on Lock & Key Model and Induced fit model. 2.5+2.5=5

F) What do you mean by enzyme? How it performs in a reaction? Write a short note on Active site, Activation energy & Enzyme specificity. 1+1+1+1+1=5

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